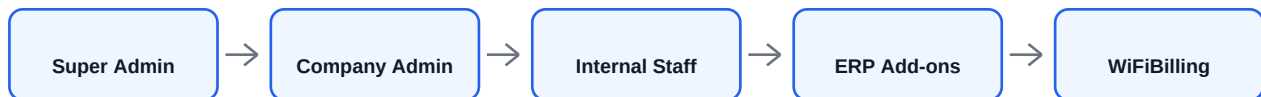


StudyRoom Connect ERP Roles, Permissions, Add-ons and Staff Assignment Guide

Implementation manual for role creation, add-on permission visibility, WiFiBilling permissions, and internal staff assignment.

Platform actor chain



Scope: Laravel ERP SaaS, Spatie Laravel Permission, Inertia React role pages, tenant-scoped staff users, and add-on-aware permission loading.

Generated for C:/xampp/htdocs/billing-system

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Executive Summary



The roles and internal staff assignment system now works for the full ERP SaaS platform, not only for WiFi Billing. Super Admin owns the platform, Company/Admin owns a tenant, and Internal Staff receive tenant-scoped roles.

The role pages render grouped permissions from the backend. The fix adds one reusable loader for create and edit, normalizes permission metadata, supports enabled add-ons, and prevents stale or incomplete grouping from hiding valid WiFiBilling rows.

Key rule: a permission must use the web guard, have usable add_on and module metadata, and pass the tenant add-on activation check before the role page can assign it.

Actor Model

Access hierarchy



Super Admin is the platform owner and should not be mixed into tenant staff assignment. Company/Admin owns a tenant and creates roles for staff. Internal Staff users receive one or more tenant-scoped roles. Add-ons such as WiFiBilling contribute permissions to the tenant.

- Super Admin: controls plans, add-ons, company tenants, and platform settings.
- Company/Admin: manages business users, roles, modules, billing, and operations.
- Internal Staff: works inside a tenant with selected permissions.
- WiFiBilling: an ERP add-on that adds ISP billing, package, router, provisioning, and voucher permissions.

Project Structure

MVC and add-on structure

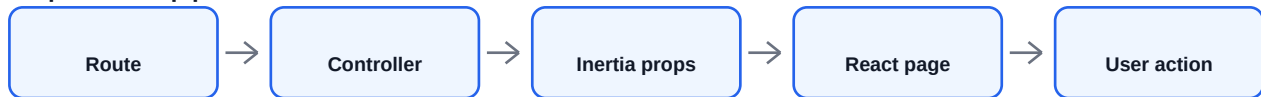


The files below are the main project areas involved in roles, permissions, add-ons, and staff assignment.

```
app/  
  Http/Controllers/RoleController.php  
  Http/Controllers/UserController.php  
  Http/Requests/StoreUserRequest.php  
  Http/Requests/UpdateUserRequest.php  
  Models/User.php  
  Models/AddOn.php  
database/  
  seeders/PermissionRoleSeeder.php  
  seeders/IspRoleSeeder.php  
  seeders/StaffRoleSeeder.php  
resources/js/pages/  
  roles/create.tsx  
  roles/edit.tsx  
  users/create.tsx  
  users/edit.tsx  
routes/  
  web.php  
packages/studyroomtechlab/WifiBilling/  
  src/Http/Controllers/DashboardController.php  
  src/Routes/web.php  
  src/Resources/js/menus/company-menu.ts
```

MVC Request Flow

Request to UI pipeline

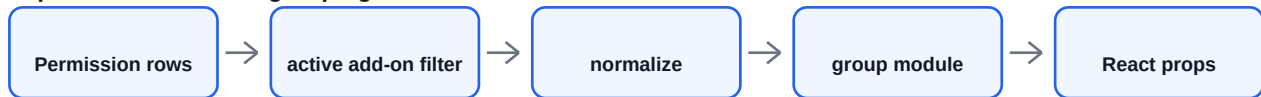


Laravel routes receive the browser request, controllers prepare tenant-scoped data, Inertia passes props to React, and React renders role or user forms. The role checkboxes are not hard-coded in the page; they are generated from the permission data sent by the controller.

- ``GET /roles/create`` maps to ``RoleController@create``.
- ``GET /roles/{role}/edit`` maps to ``RoleController@edit``.
- ``GET /users`` maps to ``UserController@index``, which passes assignable roles to user modals.
- ``POST /users`` maps to ``UserController@store``, which creates the user and assigns the selected role.

RoleController Permission Loading

Implemented backend grouping



`RoleController@create` and `RoleController@edit` now call the same reusable assignable permission loader. The loader queries Spatie permissions directly, keeps only the web guard, filters add-ons by tenant activation, normalizes missing metadata, and groups the result by add-on then module.

This avoids the old behavior where role pages depended only on `Auth::user()->getAllPermissions()`. A company/admin can now assign valid permissions for enabled add-ons even when the add-on key needs normalization.

```
$permissions = Permission::query()
->where('guard_name', 'web')
->get()
->filter(fn ($permission) => $this->permissionIsAssignable($permission))
->map(fn ($permission) => [
    'id' => $permission->id,
    'name' => $permission->name,
    'label' => $permission->label ?: Str::headline($permission->name),
    'module' => $permission->module ?: 'General',
    'add_on' => $permission->add_on ?: 'Core',
    'guard_name' => $permission->guard_name,
])
->groupBy('add_on')
->map(fn ($items) => $items->groupBy('module'));
```

Root Cause: Why WiFiBilling Was Hidden

Visibility requirements



The concrete mismatch was add-on naming. The package module key is ``WifiBilling``, while existing permission rows used ``WiFiBilling``. A strict ``Module_is_active($addOn)`` check can therefore reject valid rows even when the database permissions exist.

The fix keeps reusable add-on logic by checking the normal module activation path first, then comparing activated module names case-insensitively. WiFi Billing remains one add-on; future add-ons can use the same loader.

- ``guard_name`` must be ``web``.
- ``add_on`` should match the registered module key, with a safe normalized fallback.
- ``module`` should be a readable module group such as ``mikrotik-routers`` or ``provisioning``.
- The loader intentionally loads assignable permissions beyond the current user's direct permission list.
- ``Module_is_active('WiFiBilling')`` must return true for the tenant.
- Spatie permission cache may need reset after manual inserts.

Reusable Permission Loader

The reusable loader is in `RoleController`. Both create and edit pages use the same grouped permission shape: add_on -> module -> permissions.

- Normalize empty `add\_on` to `Core`.
- Normalize empty `module` to `General`.
- Use `label` for display and `name` for submitted value.
- Include core permissions and active add-on permissions.
- Preserve add-on grouping so the UI can render tabs.

```
private function assignablePermissions()
{
    return Permission::query()
        ->where('guard_name', 'web')
        ->get()
        ->filter(fn ($permission) => $this->permissionIsAssignable($permission))
        ->map(fn ($permission) => [
            'id' => $permission->id,
            'name' => $permission->name,
            'label' => $permission->label ?: Str::headline($permission->name),
            'add_on' => $permission->add_on ?: 'Core',
            'module' => $permission->module ?: 'General',
        ])
        ->groupBy('add_on')
        ->map(fn ($items) => $items->groupBy('module'));
}
```

Role Create UI

Create role UI structure

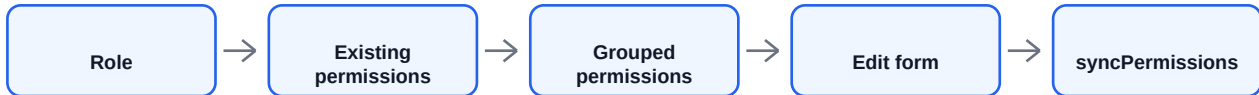


`resources/js/pages/roles/create.tsx` renders add-on tabs and module cards from the `permissions` prop. It now supports permission search plus select-all actions at all, add-on, and module levels.

- Search across add-on alias, module name, permission label, and permission name.
- Select All applies to all currently visible permissions.
- Select Add-on applies to the current tab.
- Keep module checkbox indeterminate state.
- Keep submitted values as permission names.

Role Edit UI

Edit role lifecycle

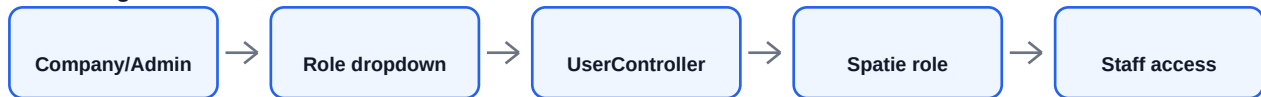


``resources/js/pages/roles/edit.tsx`` mirrors create behavior and preloads ``rolePermissions``. Because create and edit share the backend loader, WiFiBilling appears consistently in both pages.

- Use the same grouped permission prop shape.
- Preserve selected permissions from ``rolePermissions``.
- Support selecting and clearing a module after existing data loads.
- Respect non-editable role name rules while still allowing allowed permission edits where intended.

Staff Assignment Model

Staff assignment flow



Internal staff should be assigned a role/profile selected by the company/admin. The wording should be neutral ERP language: `Select a role/profile for this staff member. Permissions can be managed from Roles.`

Do not create a second role system. This project already uses Spatie Laravel Permission through `HasRoles`, `assignRole`, and `syncRoles`.

- Do not show Super Admin role in tenant staff forms.
- Load roles where `created\_by = creatorId()`.
- Validate that the selected role belongs to the same tenant.
- Use `assignRole` on create and `syncRoles` on edit.

User Creation Flow

Create internal staff sequence

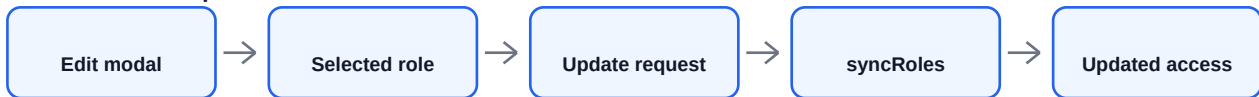


`UserController@store` now reads the staff role/profile from `role\_id` while keeping the older `type` fallback for compatibility. It validates tenant ownership, creates the user, and calls `\$user->assignRole(\$role)`.

- Validate `role\_id` and then verify it belongs to the current tenant.
- Create the user with `created\_by = creatorId()` and `creator\_id = Auth::id()`.
- Set staff type from selected role or a stable internal type such as `staff`.
- Call `\$user->assignRole(\$role)` after save.

User Edit Flow

Edit staff role sequence



`resources/js/pages/users/edit.tsx` now includes the current staff role and provides the same Role/Profile dropdown. `UserController@update` uses `\$user->syncRoles([\$role])` when a valid tenant role is selected.

- Extend `EditUserData` with `role\_id`.
- Pass the current assigned role from `UserController@index`.
- Validate the id format in `UpdateUserRequest` and tenant ownership in the controller.
- Use `\$user->syncRoles([\$role])` for a single selected role.
- Keep Super Admin roles hidden and blocked on the backend.

Database and Model Relationships

Permission model stack



The project uses Spatie Laravel Permission. Users get roles through `model_has_roles`, and roles get permissions through `role_has_permissions`. The `User` model includes the `HasRoles` trait.

- `users.created_by` scopes a user to a tenant.
- `roles.created_by` scopes a role to a tenant.
- `permissions.add_on` groups permissions by add-on.
- `permissions.module` groups permissions inside each add-on.
- `user_active_modules` and plan modules affect add-on availability.

Seeded Staff Profiles

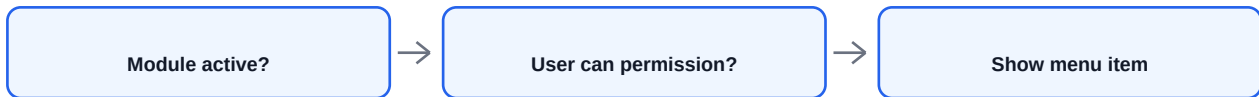
Seed common tenant roles safely. Use ``updateOrCreate`` or ``firstOrCreate`` with ``created_by`` so re-running the seeder does not duplicate roles.

- Admin
- Support Agent
- Billing Officer
- Sales / CRM
- Network Technician
- Installer
- Marketing
- Inventory Manager
- Viewer

```
Role::updateOrCreate(
    ['name' => 'network-technician', 'guard_name' => 'web', 'created_by' => $companyId],
    ['label' => 'Network Technician', 'editable' => true]
);
```

Menu Visibility

Menu decision



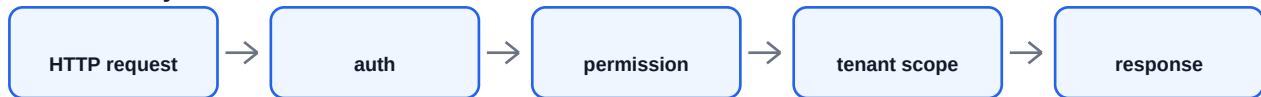
Menu visibility is controlled by both add-on activation and permission checks. Empty parent menu groups are removed after child filtering so staff do not see blank sections.

```
if (Module_is_active('WiFiBilling') && Auth::user()->can('view-mikrotik-routers')) {  
    // show MikroTik menu item  
}
```

- Company/Admin with full WiFiBilling permissions sees the module.
- Network Technician may see routers and provisioning only.
- Billing Officer may see packages, customers, invoices, and vouchers.
- Viewer may only see read-only dashboards.

Route Protection

Route security



Do not rely on React to protect data. WiFi Billing management controllers now reject unauthorized staff users while preserving tenant ownership checks.

```
public function index()
{
  abort_unless(Auth::user()->can('view-mikrotik-routers'), 403);

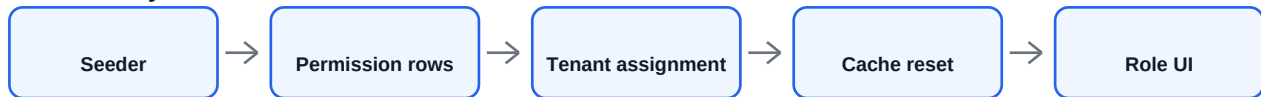
  return MikrotikRouter::where('created_by', creatorId())->get();
}
```

- Provisioning list/detail checks use `view-provisioning-logs` or `manage-provisioning`.
- Provisioning generate uses `create-provisioning-token` or `manage-provisioning`.
- Provisioning deactivate uses `deactivate-provisioning-token` or `manage-provisioning`.

`/provision/{token}` remains public/token-based and was not moved into the authenticated permission checks.

Future Add-on Permission Steps

Add-on lifecycle



Every future add-on should provide permissions through a repeatable installation/seeding path. The role pages should not need hard-coded edits for each add-on.

- Create add-on permission seeder.
- Set ``guard_name = web``.
- Set ``add_on`` to the exact module key.
- Set ``module`` to the feature group.
- Set a clear ``label``.
- Assign permissions to company/admin when the add-on is enabled.
- Clear permission cache.
- Confirm role create/edit render the new add-on tab.

Troubleshooting

Permission visibility decision tree



Use this checklist when permissions exist in the database but are not visible in the role page.

- Query the row and verify `name`, `guard\_name`, `label`, `add\_on`, and `module`.
- Check whether the current company/admin has the permission.
- Check `Module\_is\_active(add\_on)` for the current user.
- Check the add-on is installed and enabled.
- Check the active plan or user active modules include the add-on.
- Clear Laravel and Spatie permission caches.
- Inspect Inertia props for `permissions` on `/roles/create`.

Cache Clearing

Cache workflow



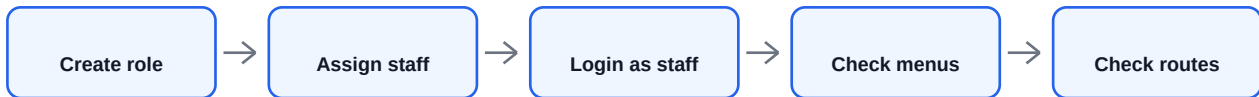
After changing permission seeders, manual permission rows, role assignments, or add-on activation, clear caches before testing.

```
C:\xampp\php\php.exe artisan optimize:clear  
C:\xampp\php\php.exe artisan config:clear  
C:\xampp\php\php.exe artisan permission:cache-reset
```

If `permission:cache-reset` is not available, continue after `optimize:clear` and `config:clear`. The command depends on the installed Spatie permission console commands.

Verification Checklist

End-to-end verification



Use this checklist after implementing the role and staff assignment fixes.

- ``/roles/create`` shows a WiFiBilling add-on group.
- ``/roles/edit`` shows the same WiFiBilling group.
- Network Technician role can be created.
- MikroTik, packages, provisioning, and voucher permissions can be selected.
- User creation shows Role/Profile, not fixed Support-only wording.
- Internal staff can be assigned Network Technician.
- Staff login only sees allowed menu items.
- Unauthorized routes return forbidden or redirect safely.
- Super Admin dashboard still works.
- ``/provision/{token}`` still works.
- PHP 8.2 syntax checks pass.

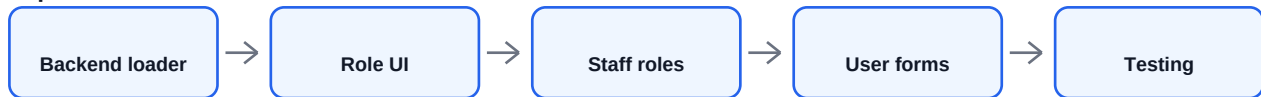
Affected Files

These files are the main places a developer should inspect or update for the roles, permissions, add-on visibility, and staff assignment work.

File path	Type	Why it matters	How to test
app/Http/Controllers/RoleController.php	Controller	Loads and groups permissions for create/edit role pages.	Open roles/create and inspect WiFiBilling groups.
resources/js/pages/roles/create.tsx	View	Renders add-on tabs, module boxes, and permission checkboxes.	Search and select permissions.
resources/js/pages/roles/edit.tsx	View	Edits existing role permissions.	Edit Network Technician.
app/Http/Controllers/UserController.php	Controller	Creates staff users and assigns selected Spatie roles.	Create staff and confirm model_has_roles.
resources/js/pages/users/create.tsx	View	Displays staff Role/Profile dropdown.	Create internal staff.
resources/js/pages/users/edit.tsx	View	Allows changing staff role.	Edit staff and sync roles.
app/Http/Requests/StoreUserRequest.php	Request	Validates selected role on create.	Submit invalid role id.
app/Http/Requests/UpdateUserRequest.php	Request	Should validate selected role on edit.	Submit role from another tenant.
database/seeder/StaffRoleSeeder.php	Seeder	Seeds default tenant staff profiles safely.	Rerun seeder in test database.
packages/studyroomtechlab/WiFiBilling	Add-on	Contributes WiFi Billing routes, menus, and permissions.	Enable add-on and open role form.

Implementation Sequence

Implemented order



These are the implemented steps and the order to repeat for another add-on or tenant role expansion.

- Step 1: Add or repair reusable permission loader.
- Step 2: Normalize permission metadata and add-on grouping.
- Step 3: Update role create/edit UI search and select-all behavior.
- Step 4: Seed staff profile roles for each company/admin safely.
- Step 5: Update user create/edit forms to use Role/Profile.
- Step 6: Validate role ownership on create and edit.
- Step 7: Assign or sync Spatie roles in UserController.
- Step 8: Clear caches and verify role pages.
- Step 9: Login as staff and confirm menus/routes.
- Step 10: Confirm Super Admin and provisioning route are untouched.